

Event Semantics

Week 6

Slides and materials based on the courses by
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A problem with verbs and adjuncts

“the gardener killed the baron”

$\mapsto \text{kill}_1(g, b)$

$\text{kill}_1 :: \langle e, \langle e, t \rangle \rangle$

“the gardener killed the baron in the park”

$\mapsto \text{kill}_2(g, b, p)$

$\text{kill}_2 :: \langle e, \langle e, \langle e, t \rangle \rangle \rangle$

“the gardener killed the baron at midnight”

$\mapsto \text{kill}_3(g, b, m)$

$\text{kill}_3 :: \langle e, \langle e, \langle e, t \rangle \rangle \rangle$

“the gardener killed the baron at midnight in the park”

$\mapsto \text{kill}_4(g, b, m, p)$

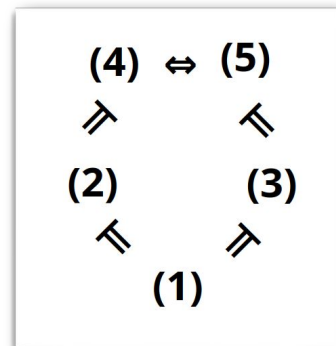
$\text{kill}_4 :: \dots$

“the gardener killed the baron in the park at midnight”

$\mapsto \text{kill}_5(g, b, m, p)$

$\text{kill}_5 :: \dots$

- Ok, but how do we explain the systematic logical entailment relations between the different uses of “kill”?



Davidson's solution: verbs as event-denoting expressions

- Verbs expressing events have an additional event argument, which is not realised at linguistic surface:
 - $\text{kill} \mapsto \lambda y \lambda x \lambda e (\text{kill}'(e, x, y)) :: \langle e, \langle e, \langle e, t \rangle \rangle \rangle$ (arity $\leftarrow$ arity + 1)
- Sentences denote sets of events:
 - $\lambda y \lambda x \lambda e (\text{kill}(e, x, y))(b)(g) \Rightarrow_{\beta} \lambda e (\text{kill}(e, g, b)) :: \langle e, t \rangle$
- **Existential closure** turns sets of events into truth conditions:
 $\lambda P \exists e (P(e)) :: \langle \langle e, t \rangle, t \rangle$
 - $\lambda P \exists e (P(e)) (\lambda e_1 (\text{kill}(e_1, g, b))) \Rightarrow_{\beta} \exists e (\text{kill}(e, g, b)) :: t$

Davisonian events and adjuncts

- Adjuncts express two-place relations between events and the respective “circumstantial information”: time, location, etc.
 - “*at midnight*” $\mapsto \lambda P \lambda e (P(e) \wedge \text{time}(e, \text{midnight}))$ $:: \langle \langle e, t \rangle, \langle e, t \rangle \rangle$
 - “*in the park*” $\mapsto \lambda P \lambda e (P(e) \wedge \text{location}(e, \text{the-park}))$ $:: \langle \langle e, t \rangle, \langle e, t \rangle \rangle$
- “*the gardener killed the baron at midnight in the park*”
 - $\mapsto \exists e (\text{kill}(e, g, b) \wedge \text{time}(e, \text{midnight}) \wedge \text{location}(e, \text{the-park}))$
 - $\models \exists e (\text{kill}(e, g, b) \wedge \text{time}(e, \text{midnight}))$
 - $\models \exists e (\text{kill}(e, g, b) \wedge \text{location}(e, \text{the-park}))$
 - $\models \exists e (\text{kill}(e, g, b))$

Compositional derivation

“the gardener killed the baron”

$$\lambda x \lambda y \lambda e [kill(e, y, x)](b)(g) \Rightarrow_{\beta} \lambda e. kill(e, g, b)$$

“... at midnight”

$$\lambda F \lambda e [F(e) \wedge time(e, m)](\lambda e_1 [kill(e_1, g, b)]) \Rightarrow_{\beta} \lambda e [kill(e, g, b) \wedge time(e, m)]$$

“... in the park”

$$\begin{aligned} & \lambda R \lambda e [R(e) \wedge loc(e, p)](\lambda e_2 [kill(e_2, g, b) \wedge time(e_2, m)]) \\ & \Rightarrow_{\beta} \lambda e [kill(e, g, b) \wedge time(e, m) \wedge loc(e, p)] \end{aligned}$$

Existential closure:

$$\begin{aligned} & \lambda P \exists e (P(e))(\lambda e_3 [kill(e_3, g, b) \wedge time(e, m) \wedge loc(e, p)]) \\ & \Rightarrow_{\beta} \exists e [kill(e, g, b) \wedge time(e, m) \wedge loc(e, p)] \end{aligned}$$

Model structures with events

- Enriched **ontological** structures
 - **Ontology**: the area of philosophy that is concerned with identifying and describing the basic “categories of being” and their relations
- A model structure with events is a triple $M = (U, E, V)$, where:
 - U is a set of “standard individuals” or “objects”
 - E is a set of events
 - $U \cap E = \emptyset$,
 - V is an interpretation function (like in first order logic)

Sorted (first-order) logic

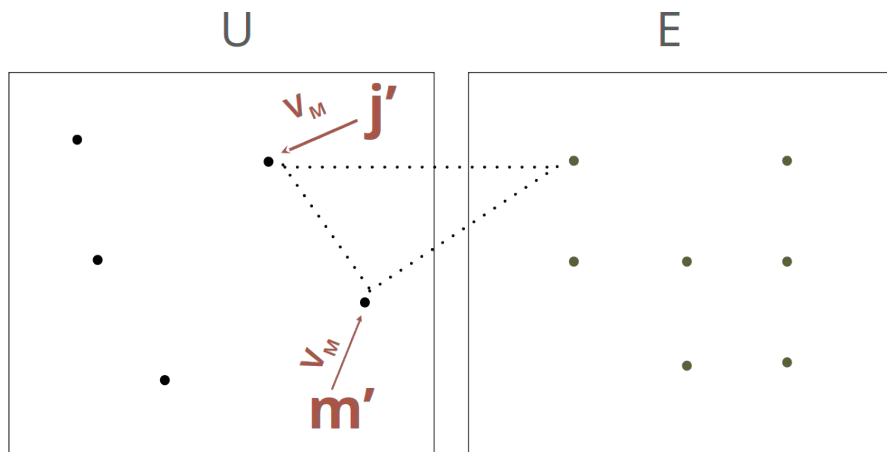
- A variable assignment g assigns individuals (of the correct sort-specific domain) to variables:
 - $g(x) \in U$ for $x \in VAR_U$ $VAR_U = \{x, y, z, \dots\}$ (Object variables)
 - $g(e) \in E$ for $e \in VAR_E$ $VAR_E = \{e, e_1, e_2, \dots\}$ (Event variables)

NB. variables from VAR_U and VAR_E are all of type e (in the formalisation used here)

- Quantification ranges over sort-specific domains:
 - $\llbracket \exists x \varphi \rrbracket^{M,g} = 1$ iff there is some $d \in U$ such that $\llbracket \varphi \rrbracket^{M,g[x/d]} = 1$
 - $\llbracket \exists e \varphi \rrbracket^{M,g} = 1$ iff there is some $s \in E$ such that $\llbracket \varphi \rrbracket^{M,g[e/s]} = 1$
 - (universal quantification analogous)

Interpreting events: example

- “John kisses Mary” $\mapsto \exists e(kiss(e, j, m))$
- $\llbracket \exists e(kiss(e, j, m)) \rrbracket^{M,g} = 1$
 - iff there is an $s \in E$ such that $\llbracket kiss(e, j, m) \rrbracket^{M,g[e/s]} = 1$
 - iff there is an $s \in E$ such that $(s, V_M(j), V_M(m)) \in V_M(kiss)$



Advantages of Davidsonian events

- Intuitive representation and semantic construction for adjuncts
- Uniform treatment of verb complements
- Uniform treatment of adjuncts and post-nominal modifiers
- Coherent treatment of tense information
- Highly compatible with analysis of semantic roles

Uniform treatment of verb complements

- “*Bill saw [an elephant]_{NP}*”
 $\mapsto \exists e \exists x(\text{see}(e, b, x) \wedge \text{elephant}(x))$ see :: $\langle e, \langle e, \langle e, t \rangle \rangle \rangle$
- “*Bill saw [an accident]_{NP}*”
 $\mapsto \exists e_1 \exists e_2(\text{see}(e_1, b, e_2) \wedge \text{accident}(e_2))$ see :: $\langle e, \langle e, \langle e, t \rangle \rangle \rangle$
- “*Bill saw [the children playing]_{VP}*”
 $\mapsto \exists e_1 \exists e_2(\text{see}(e_1, b, e_2) \wedge \text{play}(e_2, \text{the-children}))$ see :: $\langle e, \langle e, \langle e, t \rangle \rangle \rangle$

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Uniform treatment of adjuncts and post-nominal modifiers

- Treatment of adjuncts as predicate modifiers, analogous to attributive adjectives:

- “red” $\mapsto \lambda F \lambda x [F(x) \wedge \text{red}^*(x)]$ $:: \langle \langle e, t \rangle, \langle e, t \rangle \rangle$
- “in the park” $\mapsto \lambda F \lambda e [F(e) \wedge \text{loc}(e, \text{the-park})]$ $:: \langle \langle e, t \rangle, \langle e, t \rangle \rangle$

- “murder in the park”:
 $\mapsto \lambda F \lambda e [F(e) \wedge \text{loc}(e, \text{the-park})](\lambda e_1. \text{murder}(e_1))$
- “fountain in the park”:
 $\mapsto \lambda F \lambda x [F(x) \wedge \text{loc}(x, \text{the-park})](\lambda y. \text{fountain}(y))$

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Classical Tense Logic: Modal Tense Logic

- “John walks” $\mapsto$ $walk(john)$
- “John walked” $\mapsto$ $P(walk(john))$
- “John will walk” $\mapsto$ $F(walk(john))$
- Syntax like in first-order logic, plus:
 - if ϕ is a WFF, then $P\phi$, $F\phi$, $H\phi$, $G\phi$ are also WFFs such that:
 - $H\phi \equiv \neg P(\neg\phi)$
 - $G\phi \equiv \neg F(\neg\phi)$
- $P\phi$: ϕ happened in the past
- $F\phi$: ϕ will happen in the future
- $H\phi$: ϕ has always been the case
- $G\phi$: ϕ is always going to be the case

Model structures with tense information

- Tense model structures are quadruples $M = (U, T, \leq, V)$ where
 - U is a non-empty set of individuals (the “universe”)
 - T is a non-empty sets of points in time
 - $U \cap T = \emptyset$
 - $\leq$ is a linear order on T
 - V is a value assignment function, which assigns to every non-logical constant α a function from T to appropriate denotations of α
- $\llbracket P\varphi \rrbracket^{M,t,g} = 1$ iff there is a $t' < t$ such that $\llbracket \varphi \rrbracket^{M,t',g} = 1$
- $\llbracket F\varphi \rrbracket^{M,t,g} = 1$ iff there is a $t' > t$ such that $\llbracket \varphi \rrbracket^{M,t',g} = 1$

Temporal relations and events

- Observation: event structure is inherently related to temporal structure
 - *“The door opened, and Mary entered the room.”*
 - *“John arrived. Then Mary left.”*
 - *“Mary left, before John arrived.”*
 - *“John arrived. Mary had left already.”*
- How can we extend event-based models with a notion of *temporal order between events*?

Temporal event structure: ordered universe of events

- A model structure with events *and* temporal precedence is defined as:
 - $M = (U, E, <, e_u, V)$, where
 - $U \cap E = \emptyset$,
 - $(- < -) \subseteq E \times E$ is an asymmetric relation (temporal precedence)
 - $e_u \in E$ is the utterance event
 - V is an interpretation function like in standard FOL
 - Notation for overlapping events: $e_1 \cdot e_2 :=$ neither $e_1 < e_2$ or $e_2 < e_1$

Tense in semantic construction

- We can represent tense inflection as an abstract tense operator reflecting the temporal location of the reported event relative to the utterance event
 - $PAST \mapsto \lambda P. \exists e [P(e) \wedge e < e_u] :: \langle \langle e, t \rangle, t \rangle$
 - $PRES \mapsto \lambda P. \exists e [P(e) \wedge e \cdot e_u] :: \langle \langle e, t \rangle, t \rangle$
- Standard function application results in integration of temporal information and binding of the event variable (i.e. replacing existential closure):
 - “*Bill is walking*” $\mapsto PRES(Bill\ walk) \mapsto PRES(\lambda x \lambda e [walk(e, x)](b))$
 $\Rightarrow_{\beta} \exists e [walk(e, b) \wedge e \cdot e_u]$
 - “*Bill walked*” $\mapsto PAST(Bill\ walk) \mapsto PAST(\lambda x \lambda e [walk(e, x)](b))$
 $\Rightarrow_{\beta} \exists e [walk(e, b) \wedge e < e_u]$

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Verbal arguments

- Verbal arguments with the same semantic role may syntactically appear in different positions:
 - *“John broke the window with a rock”*
 - *“a rock broke the window”*
 - *“the window broke”*
- ... and we're back to the same entailment issue:

$$\exists e(\text{break}_3(e, \textcolor{red}{j}, \textcolor{blue}{w}, \textcolor{green}{r})) \not\models \exists e(\text{break}_2(e, \textcolor{green}{r}, \textcolor{blue}{w})) \not\models \exists e(\text{break}_1(e, \textcolor{blue}{w}))$$

Semantic/Thematic roles

“John broke the window with a rock”
(agent) (patient) (instrument)

“a rock broke the window”
(instrument) (patient)

“the window broke”
(patient)

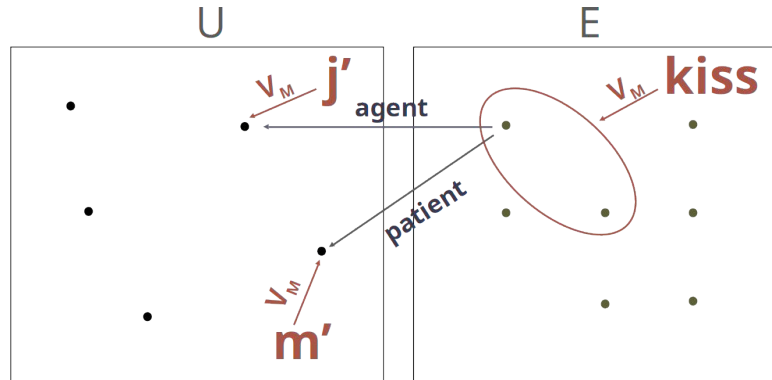
- In standard FOL & Type Theory: thematic roles are implicitly represented by the canonical order of the arguments
- In Neo-Davidsonian event semantics: thematic roles are two-place relations between the event denoted by the verb, and an argument role filler

Semantic/Thematic roles

- “John broke the window with a rock”
 $\mapsto \exists e[\text{break}(e) \wedge \text{agent}(e, j) \wedge \text{patient}(e, w) \wedge \text{instrument}(e, r)]$
- “a rock broke the window”
 $\mapsto \exists e[\text{break}(e) \wedge \text{patient}(e, w) \wedge \text{instrument}(e, r)]$
- “the window broke”
 $\mapsto \exists e[\text{break}(e) \wedge \text{patient}(e, w)]$

Interpreting thematic roles: example using event model

- “John kisses Mary” $\mapsto \exists e(\text{kiss}(e) \wedge \text{agent}(e, j) \wedge \text{patient}(e, m))$
- $\llbracket \exists e(\text{kiss}(e) \wedge \text{agent}(e, j) \wedge \text{patient}(e, m)) \rrbracket^{M,g} = 1$
 - iff there is an $s \in E$ s.t. $\llbracket \text{kiss}(e) \rrbracket^{M,g[e/s]} = 1$ and $\llbracket \text{agent}(e, j) \rrbracket^{M,g[e/s]} = 1$ and $\llbracket \text{patient}(e, m) \rrbracket^{M,g[e/s]} = 1$
 - iff there is an $s \in E$ s.t. $s \in V_M(\text{kiss})$ and $(s, V_M(j)) \in V_M(\text{agent})$ and $(s, V_M(m)) \in V_M(\text{patient})$



Verbal differences and similarities: thematic roles

- Different verbs allow different thematic role configurations:
 - a. “John broke the window with a rock” (agent, patient, instrument)
 - b. “John smiled at Mary” (agent, recipient)
 - a. “the window broke” (allows inanimate subject)
 - b. “the bread cut” (does not allow inanimate subject)
- Thematic roles capture equivalences and entailment relations between different predicates:
 - a. “Mary gave Peter the book”
 - b. “Peter received the book from Mary”

$$\forall e[\text{give}(e) \leftrightarrow \text{receive}(e)] \\ \models (a) \leftrightarrow (b)$$

Determining the role inventory

“Thematic roles form a small, closed, and universally applicable inventory conceptual argument types.” (Filmore, 1968)

- A typical role inventory might consist of the roles:
 - **Agent, Patient, Theme, Recipient, Instrument, Source, Goal, Beneficiary, Experiencer**
- But what about the following examples?
 - *“Lufthansa is replacing its 737s with Airbus 320s”*
 - *“John sold the car to Bill for 3,000€”*
 - *“Bill bought the car from John for 3,000€”*

Semantic corpora with thematic roles

- **PropBank** (Palmer et al. 2005): Annotation of Penn TreeBank with **predicate-argument structure**; separate role inventory for every lemma

- “[Lufthansa]_{ARG0} *is replacing* [its 737s]_{ARG1} *with* [Airbus A320s]_{ARG2}”
- “[Lufthansa]_{ARG0} *is substituting* [Airbus A320s]_{ARG1} *for* [its 737s]_{ARG2}”

Pred	replace
Arg0	Lufthansa
Arg1	its737s
Arg2	AirbusA320s

Pred	substitute
Arg0	Lufthansa
Arg1	AirbusA320s
Arg2	its737s

- **FrameNet** (Baker et al. 1998): A database of **frames** and a lexicon with frame information; a frame is a structured schema representing complex prototypical situations, events, and actions

- “[Lufthansa]_{AGENT} *is replacing* _{Frame:REPLACING} [its 737s]_{OLD} *with* [Airbus A320s]_{NEW}”
- “[Lufthansa]_{AGENT} *is substituting* _{Frame:REPLACING} [Airbus A320s]_{NEW} *for* [its 737s]_{OLD}”

Frame	REPLACING
Agent	Lufthansa
Old	its737s
New	AirbusA320s

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A problem with events and quantification: the interaction between quantifiers and existential closure

- “John kissed Mary”

$$\mapsto \lambda e[kiss(e) \wedge agent(e, j) \wedge patient(e, m)]$$

$$\Rightarrow_{E-CLOS} \exists e[kiss(e) \wedge agent(e, j) \wedge patient(e, m)]$$

- “John kissed every girl”

$$\mapsto \lambda e[\forall x(girl(x) \rightarrow kiss(e) \wedge agent(e, j) \wedge patient(e, x))]$$

$$\Rightarrow_{E-CLOS} \exists e[\forall x(girl(x) \rightarrow kiss(e) \wedge agent(e, j) \wedge patient(e, x))]$$

Two solutions to the event quantification problem

- **Solution I** (Champollion, 2010; 2015):
 - Interpret sentences as generalized quantifiers over events: $\langle \langle v, t \rangle, t \rangle$ instead of $\langle e, t \rangle$ (E-CLOS part of lexical semantics)
 - “kiss” $\mapsto \lambda F :: \langle v, t \rangle. \exists e (kiss(e) \wedge F(e)) :: \langle \langle v, t \rangle, t \rangle \approx \{ F \mid F \cap KISS \neq \emptyset \}$
- **Solution II** (Winter & Zwarts, 2011; de Groote & Winter, 2014):
 - Introduce separate types for regular NPs and quantified NPs, and restrict existential closure to regular NPs

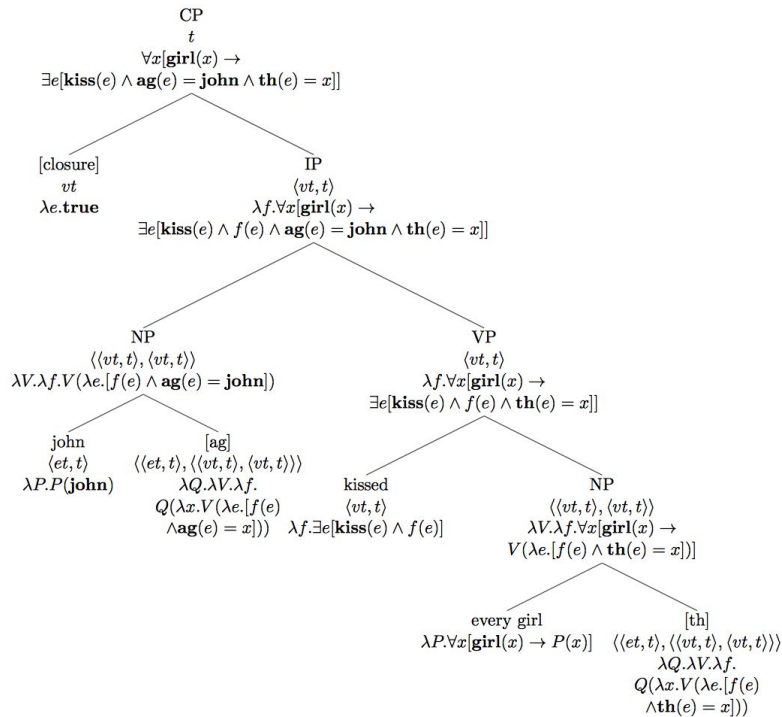
“John” $\mapsto j :: e$

“kiss” $\mapsto \lambda x \lambda y \lambda e. kiss(e, x, y) :: \langle e, \langle e, \langle v, t \rangle \rangle \rangle$

“every girl” $\mapsto \lambda Q. \forall x (girl(x) \rightarrow Q(x)) :: \langle \langle e, t \rangle, t \rangle$

E-CLOS $\mapsto \lambda P. \exists e (P(e)) :: \langle \langle v, t \rangle, t \rangle$

Solution I: Sentences as GQs over events



Solution II: Type-restriction for existential closure

$$\frac{\vdash \text{EVERY} : N \rightarrow (NP \rightarrow S) \rightarrow S \quad \vdash \text{GIRL} : N}{\vdash \text{EVERY GIRL} : (NP \rightarrow S) \rightarrow S} \quad (1)$$

$$\frac{\frac{\vdash \text{KISSED} : NP \rightarrow NP \rightarrow V \quad x : NP \vdash x : NP}{x : NP \vdash \text{KISSED } x : NP \rightarrow V} \quad \vdash \text{JOHN} : NP}{x : NP \vdash \text{KISSED } x \text{ JOHN} : V} \quad (2)$$

$$\frac{\vdash \text{E-CLOS} : V \rightarrow S \quad \frac{\vdots \quad (2)}{x : NP \vdash \text{KISSED } x \text{ JOHN} : V}}{x : NP \vdash \text{E-CLOS} (\text{KISSED } x \text{ JOHN}) : S} \quad (3)$$

$$\vdash \lambda x. \text{E-CLOS} (\text{KISSED } x \text{ JOHN}) : NP \rightarrow S$$

$$\frac{\frac{\vdots \quad (1)}{\vdash \text{EVERY GIRL} : (NP \rightarrow S) \rightarrow S} \quad \frac{\vdots \quad (3)}{\vdash \lambda x. \text{E-CLOS} (\text{KISSED } x \text{ JOHN}) : NP \rightarrow S}}{\vdash \text{EVERY GIRL} (\lambda x. \text{E-CLOS} (\text{KISSED } x \text{ JOHN})) : S}$$